The Kicked Top in Three Regimes

Classical phase space, a qubit realisation and its time reversal

Abstract

The same kicked top is studied in three regimes, regular, mixed and chaotic, selected by the kick angle p at fixed torsion strength k. Classically, the regimes are read off from Poincaré sections and explained with a small-p reduction of the map to the standard map. Quantum mechanically, the top of spin $j = n_q/2$ is realised by n_q qubits: the torsion J_z^2 is a static all-to-all Ising coupling $\sum_{i,l} V_{il}\sigma_i^z\sigma_l^z$ with $V_{il} = 1/4$, acting during the whole period, and the kick is a pulse that rotates every qubit about y. Running the chain backwards gives an exact time reversal. The regimes are compared through the Husimi distribution, the fidelity and the entanglement of a single qubit.

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1 The model

A spin $\boldsymbol{J} = (J_x, J_y, J_z)$ of length quantum number j (so $\boldsymbol{J}^2 = j(j+1)$, $\hbar = 1$) undergoes a continuous torsion about the z axis and is kicked once per unit time by a rotation about the y axis:

$$H(t) = \frac{k}{2j}\,J_z^2 + p\,J_y \sum_{n=-\infty}^{\infty} \delta(t-n). \tag{1}$$

Here k is the torsion strength per period, p the kick angle and δ the Dirac delta. Integrating the Schrödinger equation over one period, starting just before a kick, the kick acts instantaneously and the free torsion then acts for unit time. One period is therefore the Floquet operator

$$U = e^{-i\frac{k}{2j}J_z^2}\,e^{-ipJ_y}, \qquad \psi(t) = U^t\psi(0), \quad t = 0, 1, 2, \ldots \tag{2}$$

where t now counts periods.

Classical limit. The classical top is the same Hamiltonian for a classical vector $\boldsymbol{J}$. All results depend only on the direction

$$\boldsymbol{s} = \frac{\boldsymbol{J}}{j} = (X, Y, Z) = (\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta), \tag{3}$$

a point on the unit sphere with polar angle θ and azimuth ϕ. The pair (ϕ, Z) is canonical, so phase space is the rectangle $-\pi < \phi \le \pi$, $-1 \le Z \le 1$ with area 4π, and areas in it are conserved. Over one period the kick first rotates $\boldsymbol{s}$ about y by p:

$$X' = X\cos p + Z\sin p, \qquad Y' = Y, \qquad Z' = -X\sin p + Z\cos p. \tag{4}$$

The free torsion then rotates it about z by the angle kZ', which leaves Z unchanged:

$$Z_{t+1} = Z', \qquad \phi_{t+1} = \phi' + kZ', \tag{5}$$

where ϕ' is the azimuth of (X', Y'). The classical map is eqs. (4) and (5). Its quantum counterpart is eq. (2), and $1/j$ plays the role of Planck's constant.

Throughout, $k = 20$ and only p is varied:

$$p = 0.02 \text{ (regular)}, \qquad p = 0.1 \text{ (mixed)}, \qquad p = 1 \text{ (chaotic)}. \tag{6}$$

The initial state is always the north pole $Z = 1$ (all spin up).

2 Classical phase space

Figure 1 shows Poincaré sections: the points (ϕ_t, Z_t) of many orbits started on a grid. The red orbit starts at the pole.

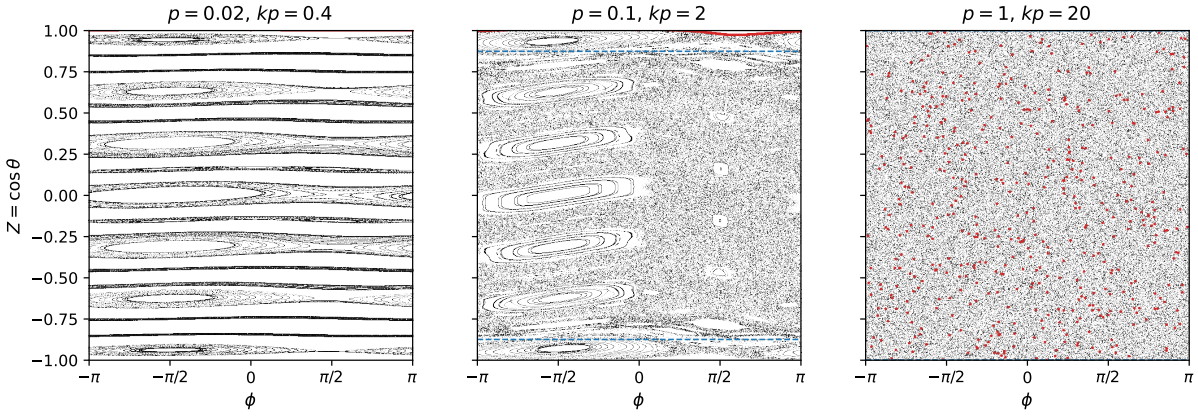


Figure 1: Poincaré sections at $k = 20$, 500 kicks of 400 orbits started on a grid (black), and the orbit started at the pole $Z = 1$ (red). Dashed: the estimate $|Z| = Z_c$ of eq. (14) for the border of the chaotic band.

2.1 Small p: the standard map

For $p \ll 1$, eq. (4) gives to first order in p, with $X = \sqrt{1-Z^2}\cos\phi$,

$$Z' = Z - p\sqrt{1-Z^2}\cos\phi. \tag{7}$$

The azimuth also changes by an amount of order p, but that is negligible next to the kick angle kZ', which is of order $k \gg 1$. Writing $I = kZ$, the map eqs. (4) and (5) becomes

$$I_{t+1} = I_t - K\cos\phi_t, \qquad \phi_{t+1} = \phi_t + I_{t+1}, \qquad K = kp\sqrt{1-Z^2}. \tag{8}$$

Z changes by at most p per kick, so K is constant over many kicks and eq. (8) is, locally in Z, Chirikov's standard map with stochasticity parameter K. ($\phi \to \phi + \pi/2$ brings it to the textbook form.) The strongest kick, $K = kp$, is at the equator, and $K \to 0$ at the poles.

Resonances. Fixed points of eq. (8) need $I_{t+1} = I_t$ up to 2π, i.e. $\cos\phi = 0$, and $\phi_{t+1} = \phi_t$ up to 2π, i.e. $I = 2\pi m$ with integer m. They lie at

$$Z_m = \frac{2\pi m}{k}, \qquad \phi = \pm\frac{\pi}{2}, \tag{9}$$

a row of resonances spaced by $2\pi/k \approx 0.31$. These are the chains of islands in the left panel of fig. 1. A small displacement $(\delta I, \delta\phi)$ from a fixed point evolves as

$$\begin{pmatrix} \delta I \\ \delta\phi \end{pmatrix}_{t+1} = \begin{pmatrix} 1 & K\sin\phi \\ 1 & 1 + K\sin\phi \end{pmatrix} \begin{pmatrix} \delta I \\ \delta\phi \end{pmatrix}_t . \tag{10}$$

The matrix has determinant 1 (the map preserves area), so its eigenvalues λ obey $\lambda^2 - T\lambda + 1 = 0$ with trace $T = 2 + K\sin\phi$. For $|T| < 2$ they are $e^{\pm i\alpha}$ and the point is stable (the centre of an island). For $|T| > 2$ one of them has $|\lambda| > 1$ and the point is unstable. Hence

$$\phi = -\tfrac{\pi}{2} :\ T = 2 - K,\ \text{stable for } 0 < K < 4; \qquad \phi = +\tfrac{\pi}{2} :\ T = 2 + K,\ \text{always unstable.} \tag{11}$$

All islands in fig. 1 sit at $\phi = -\pi/2$, and all chaos starts near $\phi = +\pi/2$.

Width of a resonance. Near $I = 2\pi m$ let $I = 2\pi m + \Delta$. For small K the map changes Δ and ϕ only a little per kick, so it is the flow $\dot\Delta = -K\cos\phi$, $\dot\phi = \Delta$ generated by the pendulum Hamiltonian $\Delta^2/2 + K\sin\phi$. The separatrix passes through the unstable point $\phi = \pi/2$, where the energy is K. It is widest at $\phi = -\pi/2$, where $\Delta^2/2 - K = K$, i.e. $\Delta = \pm 2\sqrt{K}$. In Z the island has the full width $4\sqrt{K}/k$, and the ratio of width to spacing is

$$\frac{4\sqrt{K}/k}{2\pi/k} = \frac{2\sqrt{K}}{\pi}. \tag{12}$$

While this ratio is small, invariant curves running all the way round in ϕ survive between neighbouring resonances (KAM curves), and an orbit cannot cross them. When neighbouring resonances overlap, these curves are destroyed and orbits wander in Z. Setting the ratio to 1 gives $K = \pi^2/4 \approx 2.5$. Secondary resonances lower the true threshold, known numerically for the standard map, to

$$K_c \approx 0.97. \tag{13}$$

2.2 The three regimes

Inserting $K = kp\sqrt{1 - Z^2}$ into this criterion gives the regimes directly. For the measurements below, the Lyapunov exponent Λ (the growth rate of the distance $d_t \sim e^{\Lambda t}$ between neighbouring orbits) was computed for a 30×30 grid of initial points. A point counts as chaotic if $\Lambda > 0.05$.

Regular, $p = 0.02$: $kp = 0.4 < K_c$ everywhere. KAM curves separate all resonances, and the motion is regular on the whole sphere: 0% of the grid points are chaotic. The orbit from the pole stays at $Z > 0.999$.

Mixed, $p = 0.1$: $kp = 2$. Chaos needs $K > K_c$, i.e. $|Z| < Z_c$ with

$$Z_c = \sqrt{1 - \left(\frac{K_c}{kp}\right)^2} \approx 0.87. \tag{14}$$

A chaotic band around the equator is bounded by regular caps around the poles (dashed lines in fig. 1). Inside the band, the islands at $\phi = -\pi/2$ survive because $K < 4$. 69% of the grid points are chaotic. The orbit from the pole is regular and stays at $Z > 0.97$.

Chaotic, $p = 1$**:** $kp = 20 \gg K_c$. No KAM curves remain, and except in thin strips at the poles $K > 4$, so the period-one islands are unstable as well. 100% of the grid points are chaotic, and the orbit from the pole fills the sphere.

For $K \gg 1$ the standard map stretches by about $K/2$ per kick, i.e. $\Lambda \approx \ln(K/2)$. Averaging over the sphere with Z uniform in $[-1, 1]$ (the canonical measure), and using $\ln\sqrt{1-Z^2} = \frac{1}{2}[\ln(1+Z) + \ln(1-Z)]$ and the symmetry $Z \to -Z$,

$$\frac{1}{2}\int_{-1}^{1} \ln\sqrt{1-Z^2}\, dZ = \frac{1}{2}\int_{-1}^{1} \ln(1+Z)\, dZ = \ln 2 - 1, \tag{15}$$

gives

$$\Lambda \approx \ln\frac{kp}{2} + \ln 2 - 1 = \ln(kp) - 1 \approx 2.0 \tag{16}$$

against the measured mean 1.89. At $p = 1$ the small-p reduction is no longer justified, so this is only an estimate.

3 Quantum top: phase-space resolution

The quantum analogue of a phase-space point is the spin coherent state

$$|\theta, \phi\rangle = e^{-i\phi J_z} e^{-i\theta J_y} |j, j\rangle, \tag{17}$$

the fully polarised state $|j, j\rangle$ ($J_z = j$) rotated to the direction (θ, ϕ). The Husimi distribution of a state ψ,

$$Q(\theta, \phi) = |\langle\theta, \phi|\psi\rangle|^2, \qquad \frac{2j+1}{4\pi}\int_{-\pi}^{\pi} d\phi \int_{-1}^{1} dZ\, Q = 1, \tag{18}$$

is the probability of finding ψ in the direction (θ, ϕ), drawn on the same (ϕ, Z) plane as the Poincaré sections.

The overlap of two coherent states whose directions make an angle γ is $|\langle\theta, \phi|\theta', \phi'\rangle|^2 = \cos^{4j}(\gamma/2) \approx e^{-j\gamma^2/2}$, so a coherent state is a spot of angular width $1/\sqrt{j}$. By eq. (18) it occupies an area $4\pi/(2j+1)$ of phase space, a "Planck cell". For $j = 5$ this is 1.1 out of 4π, i.e. only $2j + 1 = 11$ cells cover the sphere. One resonance strip of eq. (9) has area $2\pi \times 2\pi/k \approx 2.0$, about two cells. The quantum top at $j = 5$ therefore resolves the three regimes only coarsely: it sees whether the state stays localised or spreads, but not the individual islands.

4 Qubit realisation

4.1 Collective spin of n_q qubits

Take n_q qubits with Pauli matrices σ_i^a ($a = x, y, z$, qubit $i = 1, \ldots, n_q$) and define

$$J_a = \frac{1}{2}\sum_{i=1}^{n_q} \sigma_i^a. \tag{19}$$

Pauli matrices of different qubits commute, and $[\sigma_i^a/2, \sigma_i^b/2] = i\epsilon_{abc}\sigma_i^c/2$, where ϵ_{abc} is the Levi-Civita symbol and c is summed over. Hence $[J_a, J_b] = i\epsilon_{abc}J_c$: $\boldsymbol{J}$ is an angular momentum. The all-up state $|\uparrow \cdots \uparrow\rangle$ has $J_z = n_q/2$, the largest possible value, so it is $|j, j\rangle$ with

$$j = \frac{n_q}{2}. \tag{20}$$

The operator U is built from the J_a alone, so it commutes with $\boldsymbol{J}^2$ and with every permutation of the qubits. A state that starts in the totally symmetric subspace, of dimension $2j + 1$ and containing $|j, j\rangle$, stays in it. Within this subspace the 2^{n_q}-dimensional chain is exactly the spin-j top. Starting from the all-up state, the chain is therefore the quantum top of section 3 with $j = 5$ ($n_q = 10$) or $j = 6$ ($n_q = 12$).

4.2 Hamiltonian of the chain

The chain has the Hamiltonian of eq. (1), with J_z^2 replaced by an Ising coupling between all pairs of qubits,

$$H(t) = \frac{k}{2j}\, H_{zz} + p\, J_y \sum_{n=-\infty}^{\infty} \delta(t-n), \qquad H_{zz} = \sum_{i,l=1}^{n_q} V_{il}\, \sigma_i^z \sigma_l^z, \tag{21}$$

where i and l run over all qubits ($i = l$ included) and V_{il} is the coupling between qubits i and l. Here the coupling is the same for all pairs, $V_{il} = 1/4$. Squaring eq. (19),

$$J_z^2 = \frac{1}{4} \sum_{i,l} \sigma_i^z \sigma_l^z, \qquad \text{so} \qquad H_{zz} = J_z^2 \quad (V_{il} = 1/4), \tag{22}$$

and eq. (21) is exactly eq. (1). On a basis state with u qubits up, $\sigma_i^z = z_i = \pm 1$ and $H_{zz} = \sum_{i,l} V_{il} z_i z_l = m^2$ with $m = u - n_q/2$. The terms with $i = l$ give $\sum_i V_{ii} = n_q/4$, a constant that only adds a global phase; they are kept so that $H_{zz} = J_z^2$ holds without any phase.

4.3 Kick: single-qubit pulses

The terms of J_y in eq. (19) commute, so the exponential factorises:

$$e^{-ipJ_y} = \prod_{i=1}^{n_q} e^{-i\frac{p}{2}\sigma_i^y}. \tag{23}$$

The kick is the same rotation by p about y, applied to every qubit independently.

4.4 Free torsion: all-to-all Ising coupling

Between two kicks only the coupling acts, for unit time:

$$e^{-i\frac{k}{2j}H_{zz}}, \qquad \frac{k}{2j} = \frac{k}{n_q}. \tag{24}$$

H_{zz} is diagonal in the basis $|b\rangle$ of up and down qubits, so this multiplies every basis state by the phase $e^{-i(k/n_q)\sum_{i,l} V_{il} z_i z_l}$. One period of the chain, starting just before a kick, is

$$U_{\text{chain}} = e^{-i\frac{k}{2j}H_{zz}} \prod_i e^{-i\frac{p}{2}\sigma_i^y}, \tag{25}$$

which for $V_{il} = 1/4$ equals U of eq. (2) exactly. Numerically, for $n_q = 1, \ldots, 14$, several p and k, and coherent initial states, the amplitudes of the chain on the symmetric states agree with those of the spin-j top to 3×10^{-13} over 50 periods.

Mean-field picture. In eq. (24), the terms that contain qubit i are $(k/n_q)\, 2V_{il}\, \sigma_i^z \sigma_l^z$ with $l \neq i$, so qubit i is rotated about z by the angle $(k/n_q) \sum_{l\neq i} \sigma_l^z$. Replacing the other qubits by their mean, $\sum_{l\neq i} \sigma_l^z \approx 2\langle J_z \rangle = n_q Z$, gives the angle kZ. This is the classical torsion of eq. (5): each qubit precesses in the field of all the others.

5 Time reversal

To run the evolution backwards one needs

$$U_{\text{chain}}^{-1} = U_{\text{chain}}^{\dagger} = \prod_i e^{+i\frac{p}{2}\sigma_i^y}\; e^{+i\frac{k}{2j}H_{zz}}. \tag{26}$$

The sign of the pulse is easily reversed, $p \to -p$. The coupling, however, is always on with a fixed sign, so $e^{+i\frac{k}{2j}H_{zz}}$ cannot be obtained by flipping it. The spectrum of H_{zz} avoids this. By eq. (22) its eigenvalues are m^2 with $m = u - n_q/2$. For n_q even all m are integers, so $e^{-i2\pi m^2} = 1$. For n_q odd all m are half-integers, $m = l + \frac{1}{2}$, so $m^2 = l^2 + l + \frac{1}{4}$ and $e^{-i2\pi m^2} = e^{-i\pi/2}$ for every basis state. Hence

$$e^{-i2\pi H_{zz}} = \gamma, \qquad \gamma = \begin{cases} 1 & n_q \text{ even}, \\ -i & n_q \text{ odd}, \end{cases} \tag{27}$$

a global phase, and letting the coupling act with $2\pi - \frac{k}{2j}$ instead of $\frac{k}{2j}$ gives

$$e^{-i(2\pi - \frac{k}{2j})H_{zz}} = \gamma\, e^{+i\frac{k}{2j}H_{zz}}. \tag{28}$$

As in the spin chain, the backward period therefore changes $\frac{k}{2j} \to 2\pi - \frac{k}{2j}$, i.e. $k \to 2\pi n_q - k$, changes the sign of p, and swaps the order of the two factors:

$$U_b = \prod_i e^{+i\frac{p}{2}\sigma_i^y}\; e^{-i(2\pi - \frac{k}{2j})H_{zz}} = \gamma\, U_{\text{chain}}^\dagger. \tag{29}$$

Then

$$U_b U_{\text{chain}} = e^{+ipJ_y}\, e^{-i(2\pi - \frac{k}{2j})H_{zz}}\, e^{-i\frac{k}{2j}H_{zz}}\, e^{-ipJ_y} = e^{+ipJ_y}\, e^{-i2\pi H_{zz}}\, e^{-ipJ_y} = \gamma, \tag{30}$$

in every regime. In the language of the top, the backward period is $k \to -k$ and $p \to -p$, achieved without changing the sign of the coupling. At the turning point, the coupling of the last forward period and that of the first backward period combine into $e^{-i2\pi H_{zz}} = \gamma$: the coupling simply stays on, and the following pulses have the opposite sign. With the coupling strength fixed, a backward period lasts $(2\pi - \frac{k}{2j})/\frac{k}{2j}$ forward periods, about 2.14 for $n_q = 10$, $k = 20$. This needs $\frac{k}{2j} < 2\pi$. The construction relies on eq. (27), which holds for $V_{il} = 1/4$; for general couplings it requires the eigenvalues of H_{zz} to differ by integers.

Protocol. Starting from $\psi_0 = |\uparrow \cdots \uparrow\rangle$, apply U_{chain} for t_r periods and then U_b for t_r periods:

$$\psi(t) = U_{\text{chain}}^t \psi_0 \quad (0 \le t \le t_r), \qquad \psi(t_r + s) = U_b^s U_{\text{chain}}^{t_r} \psi_0 = \gamma^s\, U_{\text{chain}}^{t_r - s} \psi_0 \quad (0 \le s \le t_r). \tag{31}$$

Two consequences follow. The state returns, $\psi(2t_r) = \gamma^{t_r}\psi_0$, so $F(2t_r) = 1$. And every quantity is mirror-symmetric about t_r: its value at $t_r + s$ equals its value at $t_r - s$, since $\psi(t_r + s) = \gamma^s \psi(t_r - s)$. Without imperfections the echo is perfect in all three regimes, so the regimes differ only in what happens between 0 and $2t_r$. Here $t_r = 30$. Numerically, $U_b = \gamma U_{\text{chain}}^\dagger$ holds to 4×10^{-14} per step for $n_q = 1, \ldots, 14$, several p and k, and coherent initial states.

Butterfly effect. With `butterfly_effect = .true.` one qubit, $i =$ `i_flip`, is flipped at t_r, between the forward and the backward leg. The flip is a π rotation about an axis perpendicular to the initial polarisation, so that a qubit still in its initial state goes from $+s$ to $-s$: $X_i = \sigma_i^x$ for polarisation along z or y, and $X_i = \sigma_i^z$ for polarisation along x. The backward leg then gives

$$\psi(t_r + s) = U_b^s X_i U_{\text{chain}}^{t_r} \psi_0 = \gamma^s\, U_{\text{chain}}^{-s} X_i U_{\text{chain}}^{t_r} \psi_0 \quad (1 \le s \le t_r), \tag{32}$$

and at the end

$$F(2t_r) = |\langle\psi_0| U_{\text{chain}}^{-t_r} X_i U_{\text{chain}}^{t_r} |\psi_0\rangle|^2 = |\langle\psi(t_r)| X_i |\psi(t_r)\rangle|^2 = \langle X_i\rangle_{t_r}^2, \tag{33}$$

since X_i is Hermitian and unitary. The echo thus measures how far qubit i has turned away from the polarisation axis at t_r. If the state has barely moved, $\langle X_i\rangle_{t_r} \approx 0$ and the flipped qubit stays orthogonal to its initial state, $F(2t_r) \approx 0$. Numerically $F(2t_r)$ equals $\langle X_i\rangle_{t_r}^2$ to 10^{-15}. The flip breaks the permutation symmetry, so the state leaves the symmetric subspace on the backward leg. The figures below are without the flip.

Measures. The fidelity

$$F(t) = |\langle\psi_0|\psi(t)\rangle|^2 \tag{34}$$

is the probability of finding the initial state again. For a state spread evenly over the $2j+1$ symmetric states it is about $1/(2j+1)$.

By permutation symmetry, every qubit has the same Bloch vector, $\langle\boldsymbol{\sigma}_i\rangle = \langle\boldsymbol{J}\rangle/j$. The reduced state of one qubit is $\rho_i = (1 + \langle\boldsymbol{\sigma}_i\rangle \cdot \boldsymbol{\sigma})/2$ with purity $\mathrm{Tr}\,\rho_i^2 = (1+r^2)/2$, where

$$r = |\langle\boldsymbol{\sigma}_i\rangle| = \frac{|\langle\boldsymbol{J}\rangle|}{j}. \tag{35}$$

Since the whole chain is in a pure state, $r = 1$ means qubit i is unentangled with the rest, and $r < 1$ means it is entangled. r is thus both the length of the classical vector $\boldsymbol{s}$ seen by the quantum state and a measure of the entanglement created by the coupling. The total spin $\langle\boldsymbol{J}^2\rangle = j(j+1) = 30$ is conserved throughout.

5.1 Results

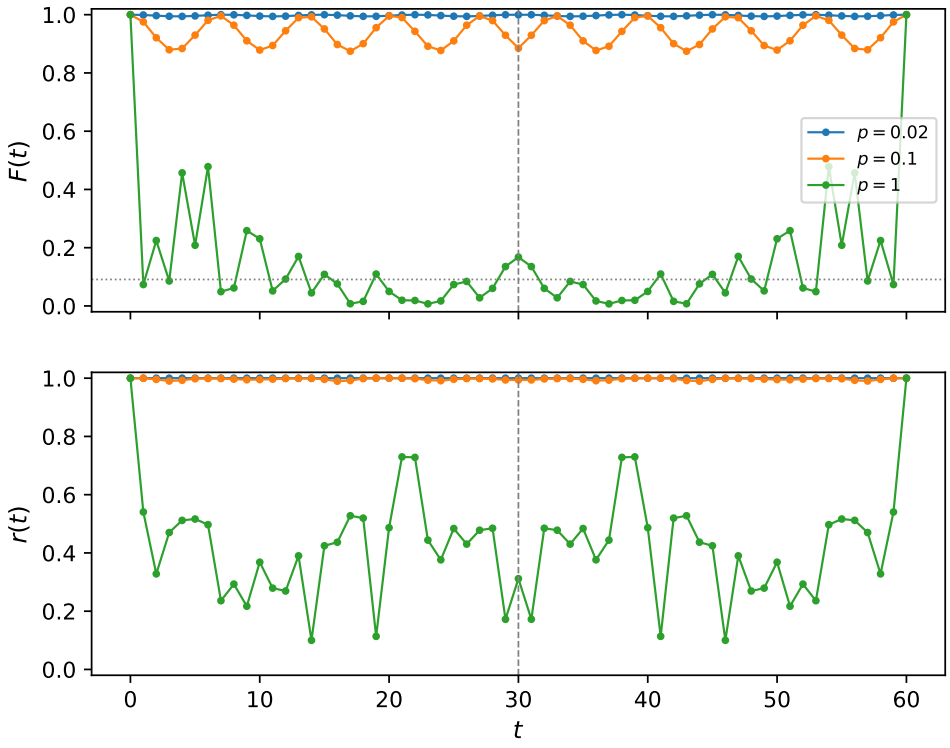


Figure 2: Fidelity $F(t)$ (top) and single-qubit Bloch-vector length $r(t)$ (bottom) for $n_q = 10$, $k = 20$, reversal at $t_r = 30$ (dashed), followed by t_r backward periods. Dotted: $1/(2j+1)$.

Regular, $p = 0.02$: The pole lies in the regular region, and the classical orbit stays within $Z > 0.999$, far inside one Planck cell. The state barely moves: $F \geq 0.994$ and $r \geq 0.99998$. The Husimi distribution at t_r is indistinguishable from the initial one.

Mixed, $p = 0.1$: The pole lies in the regular polar cap $Z > Z_c$. The state stays localised there but oscillates around the regular orbit: F oscillates between 0.87 and 1, and $r \geq 0.989$, so almost no entanglement is created. The Husimi distribution at t_r is deformed but stays in the cap.

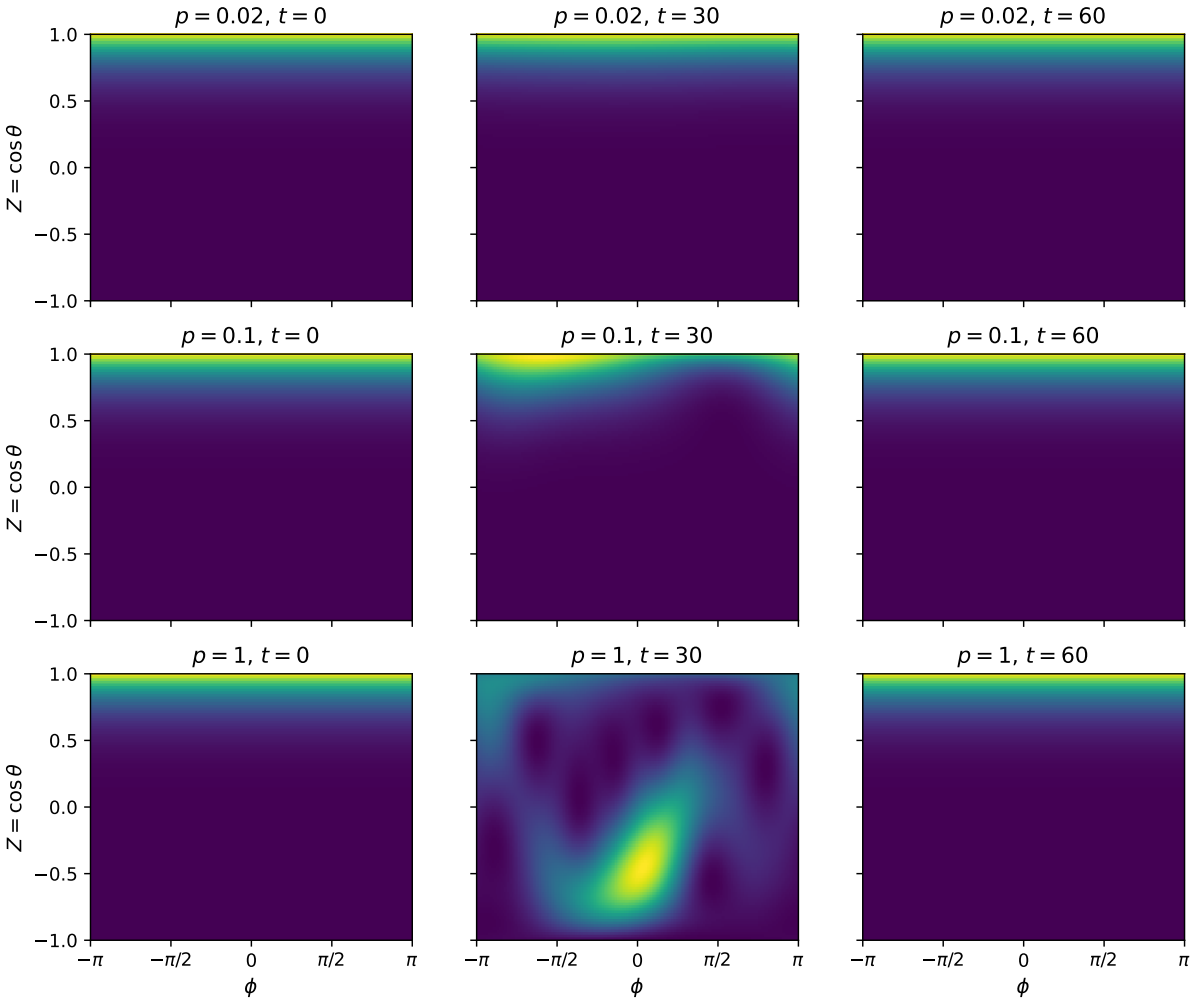


Figure 3: Husimi distribution eq. (18) of the chain state at $t = 0$, t_r and $2t_r$, for the three values of p.

Chaotic, $p = 1$: The state spreads after one kick, with $F(1) = 0.07$. This agrees with the time over which a coherent state follows the classical orbit, about $\ln j/(2\Lambda) \approx 0.4$ kicks. Afterwards F fluctuates around $0.1 \approx 1/(2j+1)$. r drops to 0.1–0.3, so each qubit is strongly entangled with the rest. At t_r the Husimi distribution is no longer a single coherent spot: its maximum drops from 0.95 to 0.45, it lies far from the pole (near $Z \approx -0.45$), and it is spread over about 5 of the 11 Planck cells, i.e. half of the sphere. (The number of cells is $(2j+1)\,(\int Q)^2/(4\pi \int Q^2)$ with $\int = \int d\phi\, dZ$, which gives about 2 for a coherent state.)

In all three regimes the backward leg mirrors the forward one, and $F(2t_r) = 1$ and $Q(2t_r) = Q(0)$ hold to 10^{-13} (figs. 2 and 3).